



# What is a Research Plan?



# The elements included in the plan

1. Background and the existing problems
  - What is the knowledge gap (thing that we don't know)?
2. Research question
  - What does this research want to find out and answer?
3. (Research goal)
  - The purpose of research. What do you want to pursue?
4. Research objective\*\*\*
  - The intended way to pursue goal and answer question.
5. Research tasks
  - The steps to achieve the objective.

# What is research?

- Research is the process of finding out something we (everyone) don't already know
- Upon completion of the work, we will have new knowledge, e.g., the result of a test of an hypothesis—false, or unfalsified
- Research is a process, not an end state—that's why there are unknowns
- Finding out something—research results in new knowledge

# RESEARCH vs. development

- If the result of your project is an artifact—a device, a product, a system, a process, an algorithm, etc.—it's probably development
- If the result of your project is new knowledge, it's probably research
- Remark: you may need development for a research, but a development may not need research

A research objective is a knowledge objective

# OBJECTIVE VS. GOAL

- Goal: To stay dry
- Objective: Replace the leaky roof over your head
- The goal is what you want to achieve, the objective is how you intend to achieve it
- Goals motivate objectives
- Objectives frame tasks
- We pursue goals, and we achieve objectives
- There may be many ways to pursue a goal, but few ways to achieve an objective

# An example

- Goal: To enable preliminary mission planning for exploration of our solar system—mission planners were finding it essentially impossible to solve the mathematical problem of space trajectory optimization
- Objective: Use a Poincaré expansion to provide an analytical solution to the calculus of variations formulation of the trajectory optimization problem for the case of finite thrust

# Stating a research objective

- Four basic ways of stating a research objective:
  - The research objective of this project is to test the hypothesis  $H$
  - The research objective of this project is to measure parameter  $P$  with accuracy  $A$
  - The research objective of this project is to prove conjecture  $C$
  - The research objective of this project is to apply method  $M$  from field  $X$  to solve problem  $Q$  in field  $Y$

# Example Hypothesis and Research Plan

- Example hypothesis: Cutting tool rake surface wear rate is a function of rake surface temperature
- Research plan:
  - Derive, from first principles, a wear model and solution
  - Plan a series of tests to cover a range of variables
  - Compare experimental results with theory

Prediction—to understand which laws of nature dominate tool wear and to formulate a predictive model.

# Example Measurement and Research Plan

- Example measurement: Measure cutting tool rake surface temperature at a point within .05 mm beneath the tool surface accurate to 10 deg. C
- Research plan:
  - Develop a temperature sensor that can be embedded in the tool
  - Fabricate test tools
  - Design and conduct a set of tests, take measurements

Data gathering—to obtain data that will enable a hypothesis test.

# Example Proof of a Conjecture and Research Plan

- Example conjecture: A bi-tangential transfer between coplanar orbits is globally optimal
- Research plan:
  - Apply Green's Theorem to reduce the dimensionality of the problem to 2 dimensions
  - Exhaustively consider all possible classes of geometries

Mathematics—to broaden our capability of thinking logically (in a self-consistent manner).

# Example Transfer of Knowledge from One Field to Another

- Example hypothesis: Apply Arrow's Impossibility Theorem (AIT) to determine the mathematical rigor of multi-scale modeling as applied to chemical-mechanical polishing (CMP)
- Research plan:
  - Extend AIT to engineering decision making
  - Apply this extension to the case of CMP

Mathematics—to resolve inconsistencies across disciplinary boundaries and broaden our ability to think consistently.

# Making the connection between the objective and research tasks

- If the research objective is properly stated, the research tasks should be obvious
- Hypothesis: the tasks define the test of the hypothesis
  - What theory/modeling is required?
  - What measurements must be taken?
  - What ranges of variables must be covered, how many data points provide a reasonable test of the hypothesis?

# Stating an hypothesis

- A typical scientific (engineering) hypothesis takes the form:  $A = B$ 
  - Mass is energy,  $E = mc^2$
  - Force is rate of change of momentum,  $F = \frac{d}{dt}(mv)$
- A typical scientific hypothesis is falsifiable
  - “Can” renders an hypothesis unfalsifiable: By adding nano particles to aluminum, I can make it stronger

# A valid hypothesis

- A valid hypothesis takes the form:  $A = B$
- A valid hypothesis is falsifiable
- For an acceptable hypothesis, the test of the hypothesis can be planned in advance of the conduct of the test
- The plan to test the hypothesis is the research plan

All our knowledge of nature is built upon unfalsified hypotheses